

Evidence-Based Guidelines for Writing Scientific Research Paper Conclusions

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Introduction

Writing effective conclusion sections requires following evidence-based guidelines that focus on synthesizing key findings, discussing implications, and providing closure without introducing new information. Strong conclusions follow established structural frameworks and serve specific rhetorical functions in scientific communication. (LLM Memory)

The conclusion section of a scientific research paper serves as the final opportunity to communicate the significance and implications of research findings to readers. Evidence-based writing guidelines emphasize that effective conclusions must synthesize the study's main results, interpret their broader meaning, and provide clear closure to the research narrative (Model-Generated). Research on scientific writing practices shows that well-crafted conclusions follow predictable structural patterns that help readers understand how individual findings contribute to the larger body of knowledge in a field (Model-Generated). Best practices indicate that conclusions should focus on answering the "so what?" question by explicitly connecting research outcomes to theoretical frameworks, practical applications, or future research directions (Model-Generated). Writing experts recommend that authors view the conclusion not merely as a summary, but as a strategic section that demonstrates the study's contribution to scientific understanding and positions the work within ongoing scholarly conversations (Model-Generated).

Structural Components and Content Requirements

Effective conclusion sections contain specific structural elements including problem restatement, key findings summary, implications discussion, study limitations, and future research directions. These components must directly address research objectives and demonstrate the study's contribution to the field. (11 sources)

Research-based guidelines identify several essential structural components that conclusion sections must include to effectively communicate research outcomes. The conclusion should begin by restating the research problem or question, though avoiding exact duplication of language from the introduction ([Akintobi et al., 2023](#)) ([Pols, 2024](#)). Following this restatement, authors must provide a concise summary of the main findings that directly correlates with the research objectives presented in the introduction ([Lustyantie et al., 2023](#)) ([Dwijayanti et al., 2024](#)) ([Piran et al., 2024](#)).

The conclusion must include interpretive content that explains how the results logically fit within the broader scientific context and how they compare with findings from other investigators ([Delvin et al., 2014](#)). Authors should provide evidence that the research problem was solved by the obtained results and include

explanatory arguments that demonstrate the study's analytical contributions ([Silva et al., 2009](#)). A critical structural requirement is the discussion of study limitations and the steps taken to address them ([Delvin et al., 2014](#)) ([Misra, 2010](#)).

The conclusion must clearly articulate the study's contribution to the overall field and explain how the work advances scientific understanding ([Okanlawon, 2018](#)) ([Lamanauskas, 2021](#)). Future research directions should be included as a distinct component, with recommendations that are concise, brief, and specific rather than broad or superficial ([Cleemput et al., 2017](#)) ([Silva et al., 2009](#)). The section should maintain clear distinctions between results, discussion, and conclusions while ensuring each research objective corresponds to a concluding statement ([Lamanauskas, 2021](#)) ([Piran et al., 2024](#)).

Evidence

(Akintobi et al., 2023)

[Writing a Good Research Paper](#)

"Writing the conclusion therefore simply entails the following (Caulfield, 2020b;Youseff, 2022): 1. Describe the research issue: In essence then Reiterating the research problem is included in the first section of the conclusion. Avoiding wording it exactly as it did in the Introduction is one thing to watch out for. It is recommended to avoid beginning your conclusion with "In conclusion" or "To conclude" given that you must restate your research problem. 2. Summarize the paper: After restating the issue, you should go back and discuss how the paper's body approached solving it and the results that resulted from that strategy. Briefly explain the main points, highlighting how each one contributed to solving the research challenge. You can also note any opposing views. Focus on describing your arguments in general terms rather than going into specifics about the data or putting out fresh ideas. Summarize your findings while making sure to make it obvious that the research challenge was resolved. 3. Talk about the implications: After summarizing your main points or conclusions, the conclusion considers the further ramifications of your study. This entails summarizing the main ideas from your article, whether they are theoretical or practical. This is frequently done by including a proposal for further study or a call to action."

(Pols, 2024)

[Elements of proper conclusions](#)

"seven key elements of effective conclusions"

"1. can be read independently from previous text; 2. restates the research question or goal; 3. reminds the reader of the study's context) show great coherence. Given the typical reader's approach first reading the abstract and, if intrigued, then the conclusions before any other sections-it is crucial that the conclusion be comprehensible independently of the other sections. The conclusion becomes independent readable if the study's context is summarized (what is the problem?) and the main research question is repeated, though not necessarily phrased as a question. Moreover, the other four elements (4. describes the method briefly; 5. answers the question (using the given context & reference to literature values if possible; 6. provides a substantiation for that conclusion; 7. provides the main recommendations and suggestions for future research)"

(Lustyantie et al., 2023)

[Application of Project Based Learning Learning to Write Scientific Articles \(Case Study of UNJ-UNNES Collaboration Class\)](#)

"The conclusion contains a brief statement of the research results obtained in accordance with the problem formulation."

(Dwijayanti et al., 2024)

[Development of Social Emotional Based Book Activity: Product Validity](#)

"The conclusion section contains a summary of the research findings, which correlate with the research objectives written in the introduction. Then state the main points of the discussion. A conclusion generally concludes with a statement about how the research work contributes to the field of study as a whole (shows how progress from the latest knowledge). A common mistake in this section is to repeat the results of an experiment, abstract, or be presented with a very list. The concluding section must provide clear scientific truths. In addition, the conclusions can also provide suggestions for future experiments."

(Piran et al., 2024)

[Enhancing Research Methodology and Academic Publishing: A Structured Framework for Quality and Integrity](#)

"Furthermore, the conclusion should encapsulate the main findings, reflect on their implications, and suggest directions for future research. In addition to reinforcing the paper's approach to the problem statement and contributing to the overall narrative, it should tie back to the introduction"

"This section summarizes the key findings and their implications, emphasizing the contribution of the research to the field. This section often reiterates the main points discussed in the paper and highlights the practical applications of the findings. Conclusions are crafted to be clear and concise, providing a final perspective on the research while suggesting directions for future investigations or applications. The conclusions must be consistent with the evidence and arguments presented throughout the paper. Each objective in the introduction should correspond to a concluding comment, allowing for a holistic perspective that aligns with the related work discussed. Additionally, authors may suggest areas for further research based on their findings in this section. A limitation of the current work can be identified; methods can be proposed to address those gaps, or topics that require further investigation can be recommended. A conclusion section highlighting future work helps readers understand where additional investigation might yield valuable insights or innovations. Key Components of the Conclusion Section are summarized as follows. • The main findings and contributions of the research should be summarized in a clear and concise manner. • The significance of the research findings and their practical applications are explained. • Consistently presenting evidence and arguments in the paper to support conclusions. • A conclusion should be written for each objective stated in the introduction. • The scope and direction of future research should be specified. • Enlightenment about the importance and potential impact of the research. • An overview of the paper in relation to the introduction and related work."

(Delvin et al., 2014)

How to Write a Scientific Paper: Practical Guidelines

"Briefly summarize the main results of your study in one or two paragraphs, and how they support your working hypothesis; . Provide an interpretation of your results and show how they logically fit in an overall scheme (biological or clinical); . Describe how your results compare with those of other investigators, explain the differences observed; . Discuss how your results may lead to a new hypothesis and further experimentation, or how they could enhance the diagnostic procedures"

"Provide the limitations of your study and steps taken to reduce them. This could be placed in the concluding remarks."

(Silva et al., 2009)

Nursing knowledge production: from the research idea to the publishing in a qualified journal

"The conclusion section can start by presenting what was learned from the study. It should be analytic, interpretative and include explanatory arguments, providing evidence that the problem/concerns were solved by the results obtained in the paper. Future research related to the same problem should be commented on, as well as necessary modifications and/or limitations of the method, which may or may not be overcome"

".When present, Final Considerations should be limited to a short explanation of the predominant ideas in the text as a whole, without great polemics or controversy, including the main considerations deriving from the results analysis. New confrontations with literature should be avoided. Depending on the research type and aim, authors can present some general recommendations for further research, raising the public's awareness about important facts, suggesting urgent decisions or more coherent practices by people or groups, with possible contributions the publication can offer to society."

(Misra, 2010)

Refining a Research Manuscript

"The discussion and conclusion section should interpret in-depth the results from the studies carried out. The researcher may begin by brief description of the problem, proposed studies that are going to be addressed and restatement of the hypothesis. The significant results should be discussed in light of prior art in the field with the references, and can be used to justify the hypothesis. The researcher may agree or disagree with the previous studies. Observed results may or may not have any similarity or connectivity to any previous studies or contrary to expectations. In such circumstances, researchers should try to justify the results with logical and scientific explanation. However, overgeneralization or superficial interpretation may be avoided in absence of authentic data or literature backing. It should also be explained that whether these studies were effective to test the hypothesis and any additional studies if required. The discussion part should also state the possibility of errors / anomalies in the results explaining its source that is, the limitations of studies. A clear-cut realistic conclusion based on proposed hypothesis should be made. It should move on to relate researchers' results to the impact it will have on the understanding and advancement of the subject under study. Conclusions can include the possible applications of the research and researchers' perception of the future direction to the work to be carried out by other researchers. No new results should crop up in discussion and conclusion section and a researcher must avoid broad statements."

(Okanlawon, 2018)

Flaws in Research Report Writing: An Evaluation of Research Reports Submitted for an International Conference on Education

"The Conclusion section introduces the work, briefly states the major results and the major points of the discussion. This aspect finally ends with a statement of how the present research contributes to the overall field of study."

(Lamanauskas, 2021)

WRITING A SCIENTIFIC ARTICLE: FOCUSED DISCUSSION AND RATIONAL CONCLUSIONS

"Clear distinction should be maintained between research results, discussion and research conclusions and implications"

"The Conclusions section usually consists of two basic parts. The first part clearly sets out the main conclusions and reveals the core and significance of the research. An important point is to reveal the methods and level used for solving the research problem and possible knowledge gaps in the researched area. Simultaneously, further directions for possible research are proposed. Conclusions may be formulated in various ways, for example, numbered or unnumbered. Nevertheless, a more appropriate format points to the conclusions presented in concise and coherent text lacking the format of theses. The formulation of conclusions focuses on seeking to answer the research questions raised. A possibly high correlation between the main purpose of the study, research questions and conclusions should be established"

"The formulation of conclusions should aim to reveal how the carried out research contributes to closing a research gap in a particular field of science and demonstrate the potential contribution of research to scientific treasure in general. Overall, conclusions must be drawn in an understandable and valid manner"

(Cleemput et al., 2017)

Manual on Scientific Communication for Postgraduate Students and Young Researchers in Technical, Natural, and Life Sciences

"Conclusions: Summarize the main results of the research in terms of the objectives given in the introduction; describe what the results mean for the respective discipline"

"The whole section ends with a summary of the important implications, regardless of the statistical significance. Provide recommendations for future research, but, keep it concise, brief, and specific."

Writing Style and Language Guidelines

Effective scientific conclusions require clear, concise language written in present tense with confident tone using active voice. Authors should avoid introducing new information while maintaining precision and authority in their writing style. (5 sources)

Evidence-based writing guidelines emphasize that conclusion sections require specific language choices and stylistic approaches to effectively communicate research outcomes. The conclusion should be written in present tense rather than past tense, as it represents updated knowledge that may be generalized, contrasting with previous sections that describe the "scientific history" in past tense ([Caceres et al., 2011](#)). This temporal shift reflects the chronological progression from describing completed research steps to presenting current contributions to knowledge.

Writing style should convey confidence and authority through the use of active voice and first person pronouns, accompanied by clarity and succinctness to convince readers of the research's value ([Kallestinova, 2011](#)). The language must be clear, precise, and concise, with authors taking responsibility to describe results and their implications in a precise manner since readers often navigate directly to conclusion sections ([Herbst et al., 2024](#)).

Effective conclusions should be short, clear, and precise, presenting only the most important findings and messages without surprising attentive readers ([Masic et al., 2014](#)). The writing should project the main message in a concise manner without mentioning information that has not been previously described in the text ([Shankar et al., 2022](#)). Authors should address potential alternative explanations proactively and suggest feasible solutions to limitations while ensuring that problems do not override the significance of findings ([Kallestinova, 2011](#)).

Evidence

(Caceres et al., 2011)

Scientific writing and the quality of papers: towards a higher impact.

"Essentially, the function of the Conclusion is to answer the proposed objectives by emphasizing the novelty found in the results. Oppositely to all the previous sections of the paper, which use the past tense, the Conclusion should be written in the present. The verbal tense must reflect the chronology of the ideas presented in the article: the "scientific history" is told in the past because it describes all the steps taken towards answering the question. The conclusion, on the other hand, should be presented in the present tense, since it represents an updated knowledge, which might or might not be generalized. In some journals, the Conclusion does

not constitute an independent section. Still, it must be stated at the end of the Discussion, respecting the same characteristics just described. In these cases, the information are usually preceded by expressions such as "In conclusion, our findings indicate that..."

(Kallestinova, 2011)

[How to Write Your First Research Paper](#)

"Another useful strategy is to be proactive in the first move by predicting and commenting on the alternative explanations of the results. Addressing potential doubts will save you from painful comments about the wrong interpretation of your results and will present you as a thoughtful and considerate researcher. Moreover, the evaluation of the alternative explanations might help you create a logical step to the next move of the discussion section: the research context. The goal of the research context move is to show how your findings fit into the general picture of the current research and how you contribute to the existing knowledge on the topic. This is also the place to discuss any discrepancies and unexpected findings that may otherwise distort the general picture of your paper. Moreover, outlining the scope of your research by showing the limitations, weaknesses, and assumptions is essential and adds modesty to your image as a scientist. However, make sure that you do not end your paper with the problems that override your findings. Try to suggest feasible explanations and solutions. If your submission does not require a separate Conclusion section, then adding another paragraph about the "take-home message" is a must. This should be a general statement reiterating your answer to the research question and adding its scientific implications, practical application, or advice. Just as in all other sections of your paper, the clear and precise language and concise comprehensive sentences are vital. However, in addition to that, your writing should convey confidence and authority. The easiest way to illustrate your tone is to use the active voice and the first person pronouns. Accompanied by clarity and succinctness, these tools are the best to convince your readers of your point and your ideas."

(Herbst et al., 2024)

[Writing an abstract](#)

"The conclusion contains the most important statement of a manuscript and usually consists of one or two sentences summarizing the main findings of the paper by addressing the hypothesis. In addition, the main results of the study should be placed in a clinically relevant context and discussed. It should be discussed how the results of the study have implications for clinical practice. As the reader often skips straight to this section of the abstract, it is the responsibility of the authors to describe the results of the study and their implications in a concise and precise manner (Andrade, 2011)."

(Masic et al., 2014)

[Ethics in Medical Research and Publication](#)

"Conclusion should be short, clear and precise. It is necessary to: Make the final statement of what logically follows from the results of the work, list only the most important and give the message. Good conclusions should not surprise attentive reader. The reader should get the impression that he himself had written it."

(Shankar et al., 2022)

[Writing Manuscripts Better: Part I \(The Introduction, Methods, Results, and Discussion Format\)](#)

"The conclusion should comprise the last paragraph of discussion, with or without a subheading. It should project the main message contained in the manuscript in a concise manner without mentioning something that has not been described previously in the text."

Relationship to Other Paper Sections

The conclusion section must maintain clear connections with all preceding paper sections, drawing directly from results while addressing the original research questions posed in the introduction. Effective conclusions create a cohesive narrative arc that demonstrates how findings answer the initial hypothesis and contribute to the broader research context. (9 sources)

The conclusion section functions as the culmination of the entire research narrative, requiring careful integration with all preceding sections of the manuscript. Most scientific manuscripts follow a standardized format where the conclusion serves as the final interpretive component that ties together the abstract, introduction, methods, results, and discussion (Evans et al., 2024). The conclusion must be directly grounded in the study's results, ensuring that all concluding statements can be traced back to empirical findings presented in the results section (Shankar, 2012).

The relationship between the conclusion and introduction is particularly critical, as the conclusion should systematically address the research questions or hypotheses originally posed. Writing experts recommend starting the writing process with the conclusion to establish a clear conceptual framework that guides the

development of all other sections, ensuring that the introduction effectively sets up the main message that will be revealed in the conclusion ([Normando, 2014](#)). This approach helps maintain focus and prevents the introduction from becoming overly broad or disconnected from the study's core findings.

The conclusion's relationship to the discussion section varies depending on manuscript structure, with some papers incorporating conclusion as a separate section while others integrate it as the final component of the discussion. When the discussion and conclusion are separate, the discussion provides the interpretive context while the conclusion emphasizes the main findings and restates the final message ([Ali et al., 2020](#)) ([Sefcik et al., 2024](#)). The discussion should evaluate whether the study hypothesis was confirmed or rejected and compare findings with other research, while the conclusion focuses on the study's relevance and implications ([Heun, 2018](#)).

The structural organization of the conclusion should mirror the logical flow established in earlier sections, with each major point in the conclusion corresponding to key elements from the introduction and results ([Courtney et al., 2016](#)). However, authors should select conclusion components based on their specific research purposes rather than including all possible elements, with some moves being obligatory while others remain optional depending on the discipline and study type ([Ahmad et al., 2021](#)). The conclusion must demonstrate how the research study effectively achieved its stated objectives, creating a clear connection between the introductory promises and the delivered outcomes ([Storey, 2023](#)).

Evidence

(Evans et al., 2024)

[Clinical Research Primer for Medical Students: Overview and Illustrative Experiences](#)

"Most manuscripts follow a standardized format: Abstract, Introduction, Methods, Results, Discussion, and Conclusion"

"The Discussion is an opportunity for the authors to interpret their data and analysis, placing the study findings within the broader context invoked by the Introduction."

(Shankar, 2012)

[What do reviewers look for in an original research article?](#)

"The conclusions should be based on the results of the study."

(Normando, 2014)

[Writing a scientific paper: Where to start from?](#)

"Start with the conclusion. To my view, one of the most common mistakes is to start writing by the introduction. How can we present something if we do not know what it is? This ordinary path, in my opinion, results from the attempt to make the best of the project that originated the study we write about. As a consequence, we end up with a long introduction that is far from conveying the main message of the research: the conclusion. Thus, starting with the conclusion infuses our mind with the draft of the path that we would like to reveal to our readers. Afterwards, it is important to keep the conclusion in mind to write the paper in a brief and linear manner."

(Ali et al., 2020)

[How to write a scientific paper? Lessons from a distinguished scientist and editor](#)

"The conclusion is where the main findings are emphasized and the final message is restated."

(Sefcik et al., 2024)

[Writing Manuscript "Bookends": Strategies to Effectively Frame Science.](#)

"The discussion ends with a brief conclusion paragraph explaining the study's relevance and implications to the field."

(Heun, 2018)

[How to write a scientific paper: A hypothesis-based approach](#)

"The discussion must state if the study hypothesis has been confirmed or rejected, if the study result is comparable to, and compatible with other research. It should evaluate the reliability and validity of the study outcome, clarify the limitations of the study"

and explore the relevance of the supported or rejected hypothesis for clinical practice and future research."

(Courtney et al., 2016)

Secrets of Successful Science Projects

"There should usually be at least one paragraph in a discussion or conclusion section of a paper for each bullet point in the discussion or conclusion section of a presentation board. This helps the student provide a logical order to their discussion, expanding on the most important points first (the direct answer to the hypothesis) and then covering less important points (limitations, confounding factors, future work, applications, new hypotheses)."

(Ahmad et al., 2021)

Generic Study of the High Impact Factor Articles of Social Sciences

"Move 1, summarizing, and Move 2, evaluating, are obligatory, Move 4, future research is conventional; however, there is a difference in Move 3. M3, practical implications are optional in social sciences research articles"

"The study concludes that these variations between the conclusion and the proposed model indicate that writers should choose the moves that best serve their purposes rather than including all the moves."

(Storey, 2023)

AI Technology and Academic Writing

"The conclusion section is again challenging to write, for while this section should be concise, it should also be detailed as it provides the core of the research completed, including recommendations and implications of the study. The formulation of conclusions should aim to reveal how the research The final formulation of conclusions shows the achieved degree of the effectiveness of the research study (introductory part, implications, recommendations)."

Common Mistakes to Avoid

Authors commonly make critical errors in conclusion sections including introducing new information, repeating results verbatim, overgeneralizing findings, and using inappropriate language or formatting. Avoiding these mistakes requires focus on synthesis rather than repetition and maintaining clear boundaries about what content belongs in conclusions. (6 sources)

Authors should avoid several critical mistakes when writing conclusion sections that can undermine the effectiveness of their scientific communication. One of the most fundamental errors is introducing new information that has no connection to the rest of the manuscript, as conclusions should only synthesize and interpret previously presented material ([Amobonye et al., 2024](#)) ([Misra, 2010](#)). Similarly, authors must avoid new confrontations with literature that were not previously discussed in the paper ([Silva et al., 2009](#)).

A common structural mistake involves repeating results verbatim rather than providing interpretive synthesis. Authors should not simply restate experimental results, repeat abstract content, or present information in list format ([Dwijayanti et al., 2024](#)) ([Yuniar, 2023](#)). Instead, conclusions should explain the most important findings and their applications rather than restating what has already been said in the findings and discussion sections ([Yuniar, 2023](#)).

Language and tone mistakes include overgeneralization or superficial interpretation when authentic data or literature backing is absent ([Misra, 2010](#)). Authors should avoid broad statements that cannot be supported by their specific findings ([Misra, 2010](#)). Additionally, conclusions should not begin with formulaic phrases like "In conclusion" or "To conclude" ([Akintobi et al., 2023](#)).

Content organization errors include failing to address research objectives systematically and not demonstrating how the research work contributes to the field as a whole ([Dwijayanti et al., 2024](#)). Authors should ensure conclusions correlate directly with research objectives written in the introduction and provide clear scientific contributions rather than vague generalizations ([Dwijayanti et al., 2024](#)). Finally, when including recommendations for future research, these should be concise and specific rather than broad or superficial ([Silva et al., 2009](#)).

Paper	Identification Of Common Mistakes	Provision Of Guidelines Best Practices	Scope Of Writing Guidance	Target Audience	Emphasis On Structure Format	Coverage Of Publication Process
Writing a Scientific Review Article: Comprehensive Insights for Beginners	Yes, the paper identifies common mistakes in titles, abstracts, the main body (e.g., excessive criticism), and general characteristics of inadequate reviews.	systematic guide for emerging scientists to write review articles	The paper provides guidance on writing and structuring publishable scientific review articles for journals.	Beginners, postgraduate students, and early-career scientists/researchers.	serves as a systematic guide for writing and structuring a review article	The paper briefly mentions high rejection rates and the importance of a good abstract for acceptance, but does not extensively cover journal selection or reviewer feedback.
Refining a Research Manuscript	Yes, the paper explicitly identifies common mistakes to avoid in scientific papers, including specific errors for the conclusion section.	Yes, the paper provides systematic instructions for writing conclusions, including content and what to avoid.	The guidance covers the entire structure and content of a standard research article/manuscript.	The primary target audience for the writing guidance is researchers.	Yes, the paper emphasizes the standard flow of information and formatting requirements for various sections of a research article.	N/A

Paper	Identification Of Common Mistakes	Provision Of Guidelines Best Practices	Scope Of Writing Guidance	Target Audience	Emphasis On Structure Format	Coverage Of Publication Process
Nursing knowledge production: from the research idea to the publishing in a qualified journal	N/A	Suggestions to improve and streamline the process	Guidance covers the entire research paper structure, specific sections, journal selection, and publication procedures.	Scientific authors, specifically in Nursing	Discusses content of components forming scientific text structure.	Yes, the paper discusses journal selection, reviewer feedback, and reasons for manuscript rejection.
Writing a Good Research Paper	Yes, the paper explicitly identifies common mistakes to avoid in the discussion and conclusion sections.	yes, it offers advice and easy actions.	Guidance for writing an entire scientific research paper, covering all major sections like Abstracts, Introduction, Literature Review, Methodology, Results and Discussion, and Conclusion.	Early researchers aiming to raise publication quality	Yes, the paper emphasizes the structural components and content requirements for specific sections of a scientific paper.	Yes, the paper discusses rejection reasons and implicitly touches on journal selection.

Evidence

(Amobonye et al., 2024)

[Writing a Scientific Review Article: Comprehensive Insights for Beginners](#)

"the author may have to justify the propositions made earlier in the manuscript, demonstrate how the paper extends past research works, and also suggest ways that the expounded theories can be empirically examined [3]. Unlike experimental studies which can only draw either a positive conclusion or ambiguous failure to reject the null hypothesis, four possible conclusions can be drawn from review articles [1]. First, the theory/hypothesis propounded may be correct after being proven from current evidence; second, the hypothesis may not be explicitly proven but is most probably the best guess. The third conclusion is that the currently available evidence does not permit a confident conclusion or a best guess, while the last conclusion is that the theory or hypothesis is false [1]. It is important not to present new information in the conclusion section which has link whatsoever with the rest of the manuscript. According to Harris et al. [90], the conclusions should, in essence, answer the question: if a reader were to remember

one thing about the review, what would it be?"

(Misra, 2010)

Refining a Research Manuscript

"The discussion and conclusion section should interpret in-depth the results from the studies carried out. The researcher may begin by brief description of the problem, proposed studies that are going to be addressed and restatement of the hypothesis. The significant results should be discussed in light of prior art in the field with the references, and can be used to justify the hypothesis. The researcher may agree or disagree with the previous studies. Observed results may or may not have any similarity or connectivity to any previous studies or contrary to expectations. In such circumstances, researchers should try to justify the results with logical and scientific explanation. However, overgeneralization or superficial interpretation may be avoided in absence of authentic data or literature backing. It should also be explained that whether these studies were effective to test the hypothesis and any additional studies if required. The discussion part should also state the possibility of errors / anomalies in the results explaining its source that is, the limitations of studies. A clear-cut realistic conclusion based on proposed hypothesis should be made. It should move on to relate researchers' results to the impact it will have on the understanding and advancement of the subject under study. Conclusions can include the possible applications of the research and researchers' perception of the future direction to the work to be carried out by other researchers. No new results should crop up in discussion and conclusion section and a researcher must avoid broad statements."

(Silva et al., 2009)

Nursing knowledge production: from the research idea to the publishing in a qualified journal

"The conclusion section can start by presenting what was learned from the study. It should be analytic, interpretative and include explanatory arguments, providing evidence that the problem/concerns were solved by the results obtained in the paper. Future research related to the same problem should be commented on, as well as necessary modifications and/or limitations of the method, which may or may not be overcome"

"When present, Final Considerations should be limited to a short explanation of the predominant ideas in the text as a whole, without great polemics or controversy, including the main considerations deriving from the results analysis. New confrontations with literature should be avoided. Depending on the research type and aim, authors can present some general recommendations for further research, raising the public's awareness about important facts, suggesting urgent decisions or more coherent practices by people or groups, with possible contributions the publication can offer to society."

(Dwijayanti et al., 2024)

Development of Social Emotional Based Book Activity: Product Validity

"The conclusion section contains a summary of the research findings, which correlate with the research objectives written in the introduction. Then state the main points of the discussion. A conclusion generally concludes with a statement about how the research work contributes to the field of study as a whole (shows how progress from the latest knowledge). A common mistake in this section is to repeat the results of an experiment, abstract, or be presented with a very list. The concluding section must provide clear scientific truths. In addition, the conclusions can also provide suggestions for future experiments."

(Yuniar, 2023)

The Hybrid Classroom's Challenges of Senior High School Students Learning

"Conclusions should not repeat what has already been said in the Findings and Discussions section but rather explain the most important findings, applications, and advances of the research conducted. This section should also demonstrate the feasibility of achieving the study objectives. Conclusions are written in the style of descriptive paragraphs. Do not use bulleted lists. Conclusions must contain contributions and recommendations, particularly for future research."

(Akintobi et al., 2023)

Writing a Good Research Paper

"Writing the conclusion therefore simply entails the following (Caulfield, 2020b;Youseff, 2022): 1. Describe the research issue: In essence then Reiterating the research problem is included in the first section of the conclusion. Avoiding wording it exactly as it did in the Introduction is one thing to watch out for. It is recommended to avoid beginning your conclusion with "In conclusion" or "To conclude" given that you must restate your research problem. 2. Summarize the paper: After restating the issue, you should go back and discuss how the paper's body approached solving it and the results that resulted from that strategy. Briefly explain the main points, highlighting how each one contributed to solving the research challenge. You can also note any opposing views. Focus on describing your arguments in general terms rather than going into specifics about the data or putting out fresh ideas. Summarize your findings while making sure to make it obvious that the research challenge was resolved. 3. Talk about the implications: After summarizing your main points or conclusions, the conclusion considers the further ramifications of your study. This entails summarizing the main ideas from your article, whether they are theoretical or practical. This is frequently done by including a proposal for further study or a call to action."

Formatting and Structural Considerations

Most journals integrate conclusions within the discussion section as a final paragraph, though some allow separate sections. Writers should select appropriate structural moves based on their research purposes and disciplinary conventions rather than following rigid templates. (5 sources)

The formatting and structural organization of conclusion sections varies significantly across journals and disciplines, requiring authors to adapt their approach based on specific publication requirements. Most journals incorporate conclusions within the discussion section rather than as a separate standalone section ([Botea et al., 2020](#)). When conclusions are integrated into the discussion, authors must include a paragraph addressing the "take-home message" that provides a general statement answering the research question and its scientific implications, practical applications, or recommendations ([Botea et al., 2020](#)).

When journals do allow separate conclusion sections, the structural format should address specific components including how the work improves the field, the practical applications of findings, identification of problems requiring solutions, and how the findings fill existing knowledge gaps ([Botea et al., 2020](#)). However, authors should recognize that not all structural elements are mandatory across all research contexts.

Research on conclusion section organization shows that certain structural moves serve different functions depending on the discipline. While summarizing findings and evaluating results are generally obligatory components, and future research directions are conventional, other elements like practical implications may be optional in some fields such as social sciences ([Ahmad et al., 2021](#)). This variation indicates that writers should choose structural moves that best serve their specific research purposes rather than including all possible components ([Ahmad et al., 2021](#)).

The formatting approach should also reflect the temporal nature of conclusions through appropriate verb tense usage. Unlike previous sections that describe completed research steps in past tense, conclusions should be written in present tense to represent updated knowledge that may be generalized ([Caceres et al., 2011](#)). When conclusions are integrated into discussion sections, they typically appear as the final paragraph and should project the main message concisely without introducing previously undescribed information ([Shankar et al., 2022](#)). The structural organization must integrate analytical elements that create persuasive arguments, a critical skill for effective scientific communication ([Kanoksilapatham, 2023](#)).

Evidence

(Botea et al., 2020)

[How to Write a Clinical Paper: Basics, Tips, Traps - Part II.](#)

"For most journals, the Conclusions are included in the Discussion section. If your submission does not require a separate Conclusion section, adding a paragraph about the "take-home message" is mandatory. This should be a general statement on your answer to the research question and its scientific implications, practical application, or advice (11). In the rare situations when separated from Discussion, describe briefly and clearly: • How your work improves the field. • Indicate the use of your work. • Which is the issue that needs a solution? • What is the current knowledge about this issue? • Are there any existing solutions to this issue? • Which are the limitations or gaps in the current knowledge about this issue? • How does your findings fill these limitations or gaps?"

(Ahmad et al., 2021)

[Generic Study of the High Impact Factor Articles of Social Sciences](#)

"Move 1, summarizing, and Move 2, evaluating, are obligatory, Move 4, future research is conventional; however, there is a difference in Move 3. M3, practical implications are optional in social sciences research articles"

"The study concludes that these variations between the conclusion and the proposed model indicate that writers should choose the moves that best serve their purposes rather than including all the moves."

(Caceres et al., 2011)

[Scientific writing and the quality of papers: towards a higher impact.](#)

"Essentially, the function of the Conclusion is to answer the proposed objectives by emphasizing the novelty found in the results."

Oppositely to all the previous sections of the paper, which use the past tense, the Conclusion should be written in the present. The verbal tense must reflect the chronology of the ideas presented in the article: the "scientific history" is told in the past because it describes all the steps taken towards answering the question. The conclusion, on the other hand, should be presented in the present tense, since it represents an updated knowledge, which might or might not be generalized. In some journals, the Conclusion does not constitute an independent section. Still, it must be stated at the end of the Discussion, respecting the same characteristics just described. In these cases, the information are usually preceded by expressions such as "In conclusion, our findings indicate that..."

(Shankar et al., 2022)

[Writing Manuscripts Better: Part I \(The Introduction, Methods, Results, and Discussion Format\)](#)

"The conclusion should comprise the last paragraph of discussion, with or without a subheading. It should project the main message contained in the manuscript in a concise manner without mentioning something that has not been described previously in the text."

(Kanoksilapatham, 2023)

[Stand-Alone Conclusion Section in Open-Access Research Articles: Organizational Structure](#)

"This study has provided insights into integrating analytical elements to successfully stage persuasive arguments in the conclusion section, a skill that needs to be inculcated in novice or early-career researchers and seasoned researchers alike."